EPIGENETICS

Module map

- Module 10: Epigenetics
- Connected lab: lab-10_epigenetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Gene regulation as control

DNA methylation

Module 10

Epigenetics versus mutation

Histone and chromatin change

Environment and expression

lab-10_epigenetics.md

EPIGENETICS

Learning objectives

- Explain gene regulation as control of gene activity.
- Describe DNA methylation at an introductory level.
- Connect chromatin structure to gene accessibility.
- Give a careful example of environmental influence on expression.
- Distinguish epigenetic regulation from mutation.

OBJECTIVE 1
Explain gene regulation as control of gene activity.

OBJECTIVE 2
Describe DNA methylation at an introductory level.

OBJECTIVE 3
Connect chromatin structure to gene accessibility.

OBJECTIVE 4
Give a careful example of environmental influence on expression.

OBJECTIVE 5
Distinguish epigenetic regulation from mutation.

EPIGENETICS

Topic sequence

- Gene regulation as control: Cells regulate which genes are active rather than using every gene equally.
- DNA methylation: DNA methylation can reduce transcription without changing base sequence.
- Histone and chromatin change: Histone and chromatin changes alter access to DNA.
- Environment and expression: Environmental conditions can influence gene expression patterns.
- Epigenetics versus mutation: Epigenetic change affects gene activity, while mutation changes DNA sequence.

TOPIC 1
Gene regulation as control

TOPIC 2
DNA methylation

TOPIC 3
Histone and chromatin change

TOPIC 4
Environment and expression

TOPIC 5
Epigenetics versus mutation

EPIGENETICS

Module 10: Epigenetics Evidence Map

- Module focus: Epigenetics
- Central claim: Epigenetics explains gene regulation through chromatin, environment, and expression evidence.
- Key nodes to connect: Gene regulation as control, DNA methylation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-10_epigenetics.md supplies an evidence surface for the map.

EPIGENETICS

Module 10: Epigenetics Reasoning Sequence

- Module focus: Epigenetics
- Trace the model: Gene regulation as control -> DNA methylation -> Histone and chromatin change -> Environment and expression
- Inputs become outputs: Gene regulation as control, DNA methylation -> Explain gene regulation as control of gene activity., Describe DNA methylation at an introductory level.
- Feedback check: lab-10_epigenetics.md evidence checks whether students can explain histone and chromatin change.
- Lab connection: lab-10_epigenetics.md gives students a process to observe or test.

EPIGENETICS

Terms as evidence handles

- Gene regulation: Control of when and how strongly genes are used.
- Epigenetics: Heritable or persistent expression changes without DNA sequence change.
- DNA methylation: Chemical marking of DNA often associated with reduced transcription.
- Histone: Protein around which DNA is packaged.
- Chromatin: DNA-protein complex that can be open or compact.
- Transcription factor: Protein that helps control transcription.

GENE REGULATION
Control of when and how strongly genes are used.

EPIGENETICS
Heritable or persistent expression changes without DNA sequence change.

DNA METHYLATION
Chemical marking of DNA often associated with reduced transcription.

HISTONE
Protein around which DNA is packaged.

CHROMATIN
DNA-protein complex that can be open or compact.

TRANSCRIPTION FACTOR
Protein that helps control transcription.

EPIGENETICS

Lab connection

- Lab file: lab-10_epigenetics.md
- Lab evidence should test or illustrate: Histone and chromatin change
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
Gene regulation as control

EVIDENCE
lab-10_epigenetics.md

CLAIM
Explain gene regulation as control of gene activity.

EPIGENETICS

Contrast check

- Do not stop at: Cells regulate which genes are active rather than using every gene equally.
- Stronger explanation: Epigenetic change affects gene activity, while mutation changes DNA sequence.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
Cells regulate which genes are active rather than using every gene equally.

DEEPER
Epigenetic change affects gene activity, while mutation changes DNA sequence.

EPIGENETICS

Module 10: Epigenetics Retrieval and Lab Check

- Module focus: Epigenetics
- Retrieval prompt: Why do cells regulate genes?
- Answer check: Explain gene regulation as control of gene activity.
- Required terms: Gene regulation, Epigenetics, DNA methylation
- Lab connection: lab-10_epigenetics.md; lab-10_epigenetics.md supplies evidence for epigenetics evidence from expression without sequence change.

EPIGENETICS

Practice quiz bridge

- 1. Which statement best defines Gene regulation in Module 10?
- 2. Which learning goal best supports the topic "DNA methylation"?
- 3. How should lab-10_epigenetics.md support Module 10?
- 4. Which retrieval move best prepares a student for Epigenetics?

QUIZ 1
A

QUIZ 2
B

QUIZ 3
C

QUIZ 4
D

EPIGENETICS

Synthesis and exit ticket

- Exit claim: explain gene regulation as control using evidence from lab-10_epigenetics.md.
- Must include: Gene regulation, Epigenetics, DNA methylation.
- Revision prompt: what evidence would change your explanation?

CLAIM
Gene regulation as control

EVIDENCE
lab-10_epigenetics.md

REVISION
What would change your mind?